

The statistics of soaring flight

Displacement scaling across
terrain, circuit types
and wing classes

Matteo Di Sante

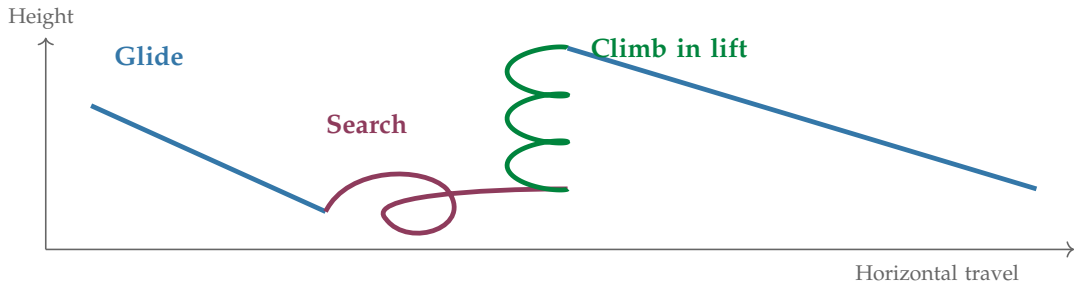
EconophysicsLab offsite 2026



Soaring is transport with an energy budget

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Height is spent to travel and recovered in rising air.



Wind, terrain and route choice act throughout the flight.

Schematic

Which statistical laws emerge from this sequence of behaviours?

A large archive of free flight

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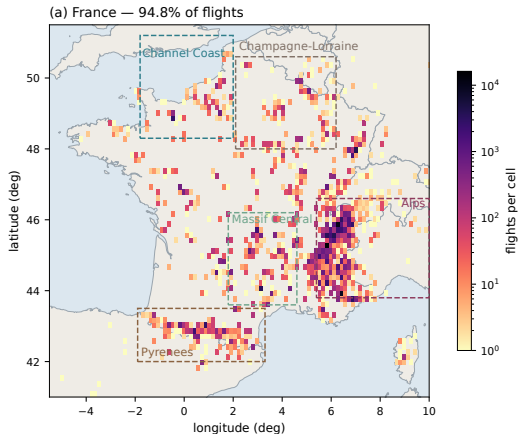
FFVL distance competition

186,052

Paraglider tracklogs
1999–2000 to 2025–2026

6,716

Hang-glider tracklogs
2001–2002 to 2025–2026



IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

Preprocessing: clean the recorded flight

4

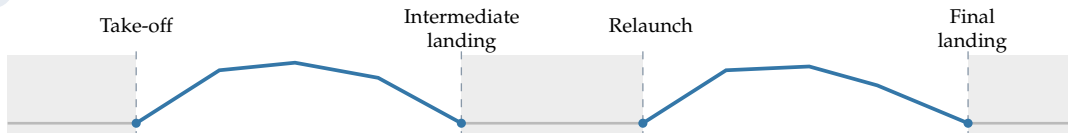
1 GNSS altitude

GNSS height $\longrightarrow$ Trajectory $z(t)$

Barometer $\longrightarrow$ Cleaning checks

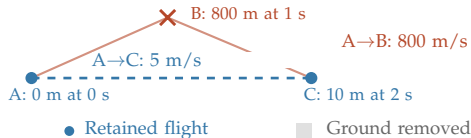
GNSS needs no pressure or temperature calibration.

3 Airborne interval



2 Fix-level cleaning

One fix: time, position and altitude.
Remove a faulty fix only when the remaining fixes can be physically rejoined.



Trim before take-off and after landing; excise witnessed stops and split the flight

4 Flight selection

Minimum flight thresholds

Blue: passes this check

Time ≥ 40 min

1.13% excluded at this check

Path ≥ 20 km

0.09% excluded at this check

Altitude range ≥ 600 m

2.68% excluded at this check

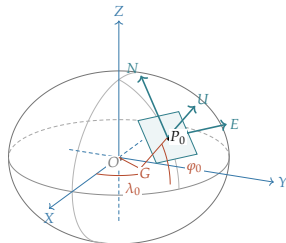
Plus integrity and reach checks

Retained after the full pipeline

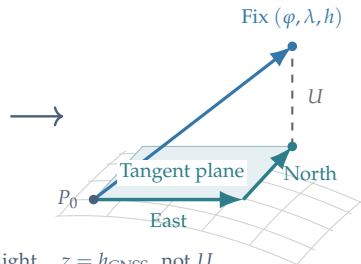
156,406 paragliders 84.1%

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred position
 $\rightarrow$ local east-north distances in metres



P₀: first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U.



6,094 hang gliders 90.7%

6 Time grid

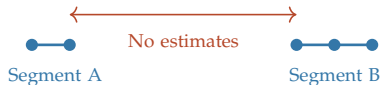
Keep the flight's native cadence Δt .
Bridge only when $g \leq g_{\max} = \min(10\Delta t, 20\text{ s})$.

Example: $\Delta t = 5\text{ s}$, $g_{\max} = 20\text{ s}$

Short gap
15 s: fill



Long gap
30 s: split

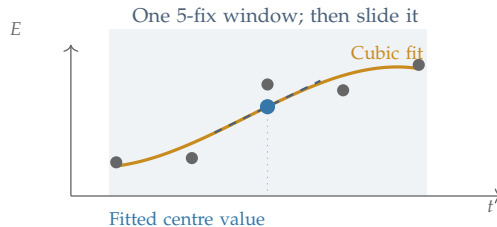


● Retained fix

○ Reconstructed fix

7 Savitzky-Golay fit

Fit a cubic locally to E , N , z .
Differentiate the polynomial analytically.



$$\widehat{x^{(n)}}(t_i) = P_i^{(n)}(0), \quad u = t - t_i$$

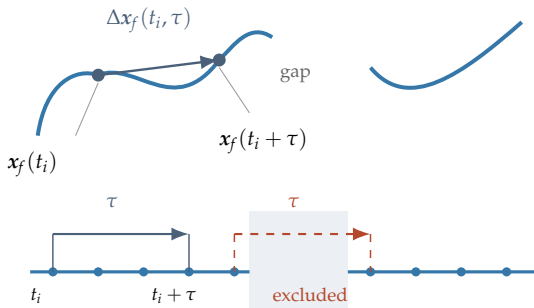
$n = 0$: position; $n = 1$: velocity; $n = 2$: acceleration

Analytic derivatives, without finite differences; fit each segment separately

Measuring horizontal transport

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One flight f , two retained segments



$\mathcal{I}_f(\tau)$: origin indices with both endpoints inside the same retained segment.

Pool origins within each flight;
then give each flight equal weight.

$$\Delta \mathbf{x}_f(t_i, \tau) = \mathbf{x}_f(t_i + \tau) - \mathbf{x}_f(t_i)$$

$$m_f(\tau) = \frac{1}{|\mathcal{I}_f(\tau)|} \sum_{i \in \mathcal{I}_f(\tau)} |\Delta \mathbf{x}_f(t_i, \tau)|^2$$

$$M_2(\tau) = \frac{1}{|\mathcal{C}(\tau)|} \sum_{f \in \mathcal{C}(\tau)} m_f(\tau)$$

$$\mathcal{C}(\tau) = \{f : |\mathcal{I}_f(\tau)| > 0\}: \text{contributing flights.}$$

$$M_2(\tau) \propto \tau^{2H}: \quad H = \frac{1}{2} \text{ diffusion} \quad H = 1 \text{ ballistic motion}$$

Why use a fixed population?

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All available flights

As lag grows, short flights drop out.
MSD then mixes motion with a change
in which flights are averaged.

C_{10000} : select once for the largest lag

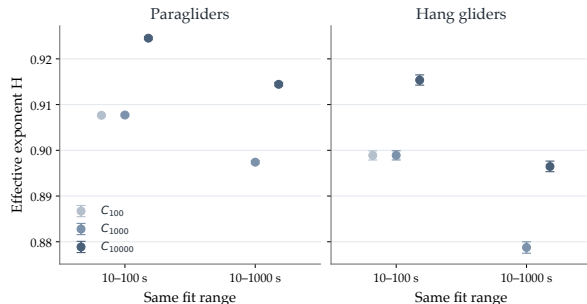
$10,000 \text{ s} \leq 0.8T_s$: $T_s \geq 12,500 \text{ s}$

Only these segments enter at every lag.

46,273 paragliders

2,326 hang gliders

Cost of selecting longer flights



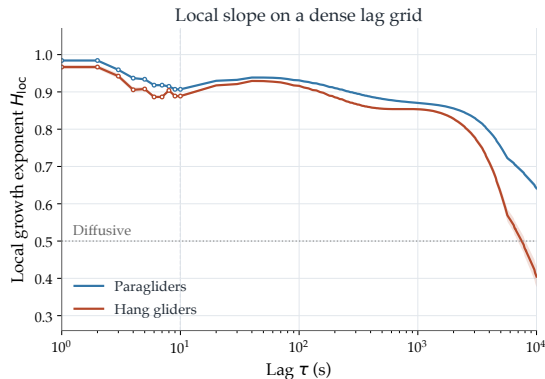
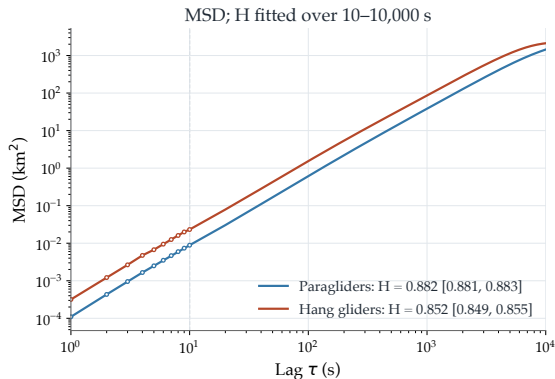
H rises by about 2% on matched lag ranges.

Fixed flights and segments over 10–10,000 s; fewer origins at longer lags

Displacement growth from 1 s to 10,000 s

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C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.



Below 10 s the sample changes, so the local slope also reflects composition.

$$H_{\text{PG}} = 0.882 [0.881, 0.883]; H_{\text{HG}} = 0.852 [0.849, 0.855] \text{ over } 10\text{--}10,000 \text{ s}$$

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Which flight characteristics shape MSD and H ?

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Circuit type

Declared route
Open versus closed

Terrain / altitude

Proxy: launch altitude
Plains, hills, low and
high mountains

Experience

Proxy: wing class
Beginners: EN A/B/C
Experts: EN D/CCC

Marginal comparison

Pool the other characteristics.
Open vs closed, all altitudes together.

Stratified comparison

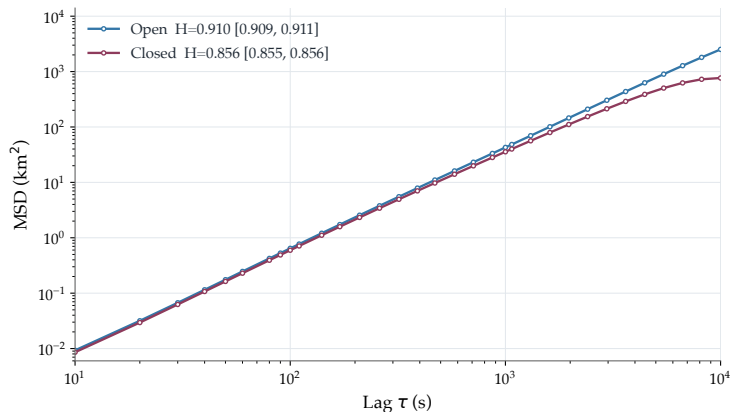
Keep a selected category fixed.
Open vs closed, one altitude class at a time.

One factor at a time (OFAT): vary one input while holding the others fixed.
The archive comparisons leave other flight conditions free to differ.

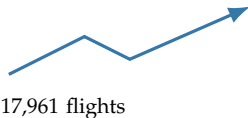
First compare groups; then test which contrasts persist across shared settings

How do MSD and H differ by circuit type?

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Open circuits



Closed circuits



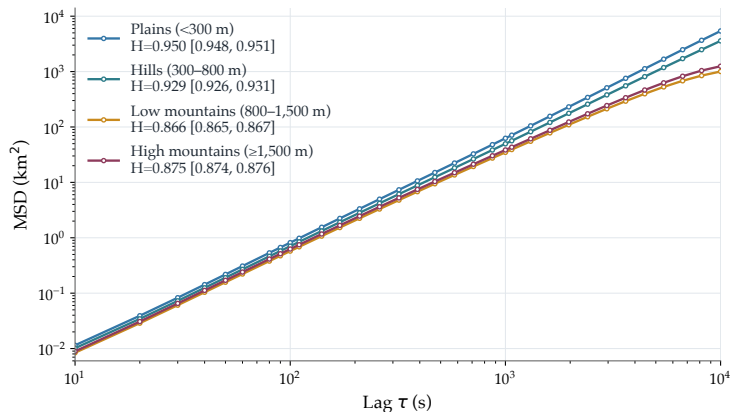
$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] \quad (90\% \text{ paired interval})$$

Slower closed-route MSD growth is consistent with turns and returns.

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

Why is pooled H higher in lowlands?

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Circuit composition

	Open	Closed
Plains	91.1%	8.9%
Hills	71.1%	28.9%
Low mtn.	33.5%	66.3%
High mtn.	35.0%	64.8%

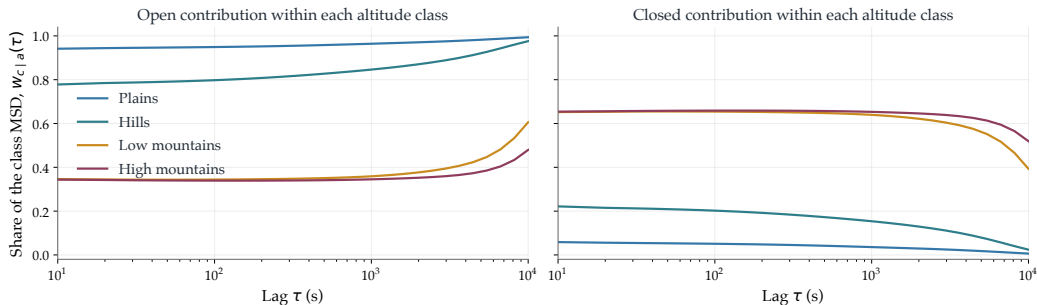
Shares within each altitude group.
Unclassified: less than 0.3%.

Plains $>$ Hills $>$ Mountains in pooled H ; lowlands have larger open fractions.
The pooled altitude contrast also reflects circuit mix.

Initial GNSS altitude above sea level. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

$$w_{c|a}(\tau) = \frac{p_{c|a} M_{2,c,a}(\tau)}{M_{2,a}(\tau)} \quad p_{c|a} = \frac{N_{c,a}}{N_a}$$

Within altitude class a ;
circuit c , flight fraction p .

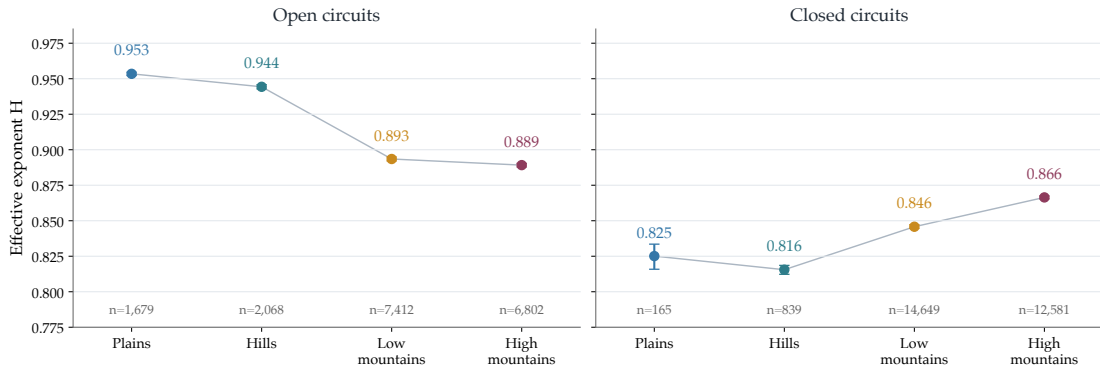


Low mountains: 33.5% open flights supply 60.6% of the MSD at 10,000 s.

Changing MSD shares shape both the amplitude and fitted H of the pooled curve.

Does the circuit contrast persist across terrain?

14



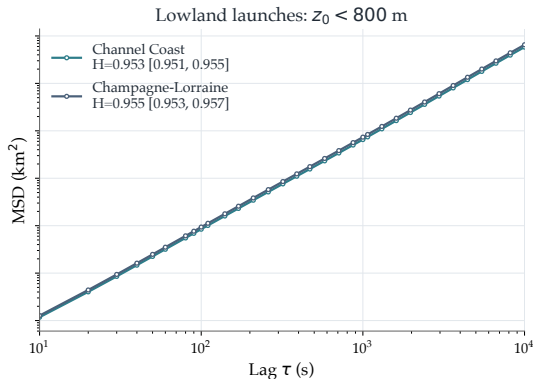
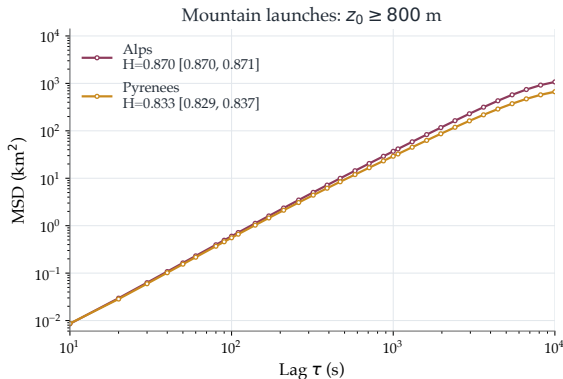
Open minus closed (ΔH , 90% CIs): Plains 0.128 [0.120, 0.138]; Hills 0.129 [0.125, 0.132].
Low mountains 0.048 [0.046, 0.049]; High mountains 0.023 [0.021, 0.024].

Open: lowland H is higher. **Closed:** mountain H is higher.
Open > closed within every altitude class; the gap narrows in mountains.

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Is launch altitude enough to describe transport?

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ΔH : Alps–Pyrenees 0.037 [0.033, 0.041]; lowlands -0.002 [-0.005 , 0.001].

Broad altitude classes do not capture all transport differences.

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

How do climb locations change with height?

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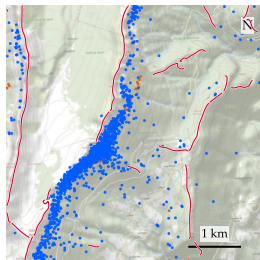
All dates and years pooled. Vilpellet climb crossings.

H2: Chartreuse, Alps

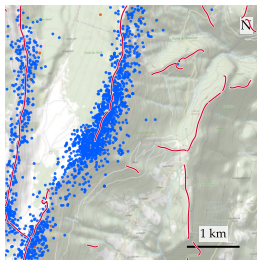
High mountains #2 / 15,840 flights

$z_{AGL} = 100\text{ m}$

$z_{AGL} = 600\text{ m}$



1,856 crossings / 1,261 flights



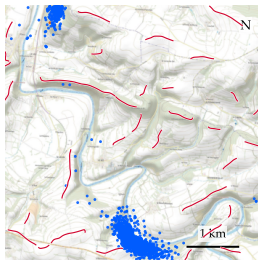
1,868 crossings / 1,452 flights

P1: Suisse Normande, Channel Coast

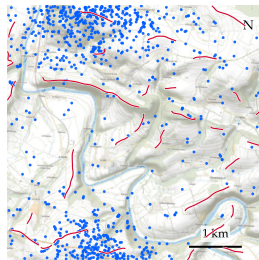
Plains #1 / 2,246 flights

$z_{AGL} = 100\text{ m}$

$z_{AGL} = 600\text{ m}$



1,441 crossings / 734 flights



901 crossings / 574 flights

100 to 600 m: P1 climbs spread out; H2 climbs stay near slopes and ridges.

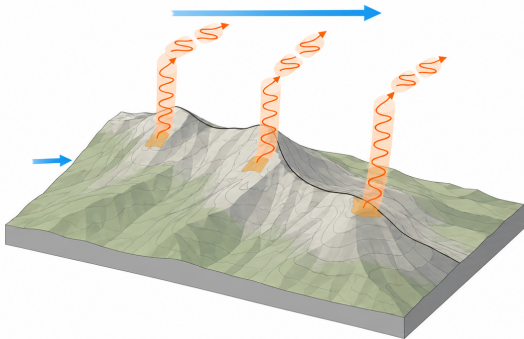
Observed lift use is less confined aloft in P1, supporting more spatial freedom for route choices.

Why do lowland climb locations spread aloft?

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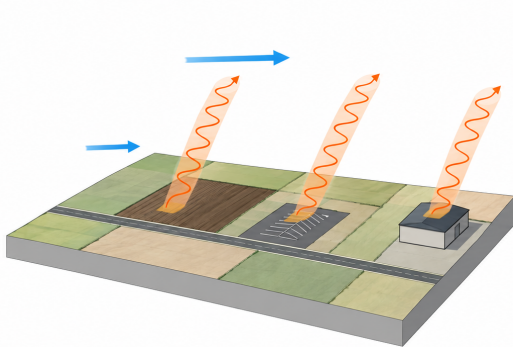
Mountains: sources on heated slopes

Possible shelter below crests; stronger shear aloft.



Lowlands: separated heated surfaces

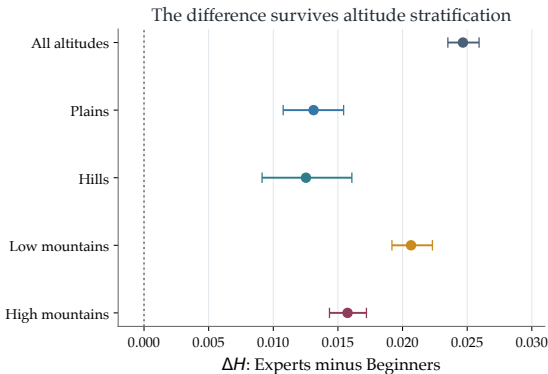
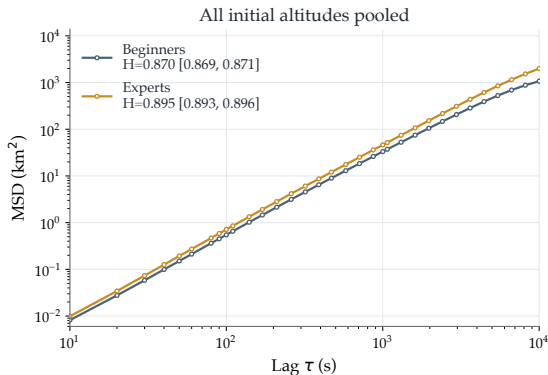
Dry field, asphalt, warm roof; drift from the base.



Distributed sources and variable wind could broaden lowland crossings.

Mountain slope sources and possible shelter can keep lift use closer to terrain corridors.

Orange: rising air; blue: wind. Illustrative profiles, not measurements. Sources: FAA, Ch. 9–10; SSA, Lift Sources.



Higher H for experts in every class; the gap depends on altitude.
A constant speed advantage changes MSD amplitude, not H .

Segmentation

Vilpellet's HMM: developed on tracks at ≥ 1 Hz.
Slower logs and some missed climbs motivate a revision.
Next: continuous observations and timing in physical seconds.

Anisotropy

Mountains: test alignment with the mountain-chain axis.
Plains: test weaker alignment with mean wind.
Preliminary observations; quantify under matched conditions.

Distribution scaling

Collapse tests and moment scaling currently give mixed indications.
A power-law MSD alone does not establish self-similarity.

Validate across logger cadences before interpreting phase statistics and scaling

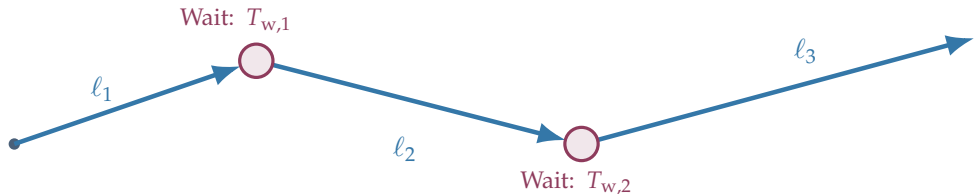
A two-phase model, motivated by Vilpellet et al.

Displacement: gliding

Duration T_g , length ℓ and direction θ .
Retain turning-angle memory between glides.

Waiting: search + thermalling

T_w : total time between successive glides.
Estimate one pooled waiting-time distribution.



Waiting neglects local search motion and thermal drift: a modelling approximation to test.

Monte Carlo: test MSD, displacement distributions, anisotropy and memory

Solo versus grouped flights

Build reliable group / solo labels.
Compare transport within shared terrain,
circuit and wing-class settings.

$$\Delta H_{a,c,e}^{\text{social}} = H_{a,c,e,\text{group}} - H_{a,c,e,\text{solo}}$$

Factorial models: estimate this contrast
and how it changes across the other factors.

Where is usable lift?

Estimate the horizontal distribution of thermals
conditional on height and weather.
Account for where gliders actually sampled.



Test routes against an explicit objective:
for example, maximise expected progress
subject to altitude and safety constraints.

Separate social contrasts from composition; link route choices to the available lift

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.

parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

[IGN / Géoplateforme](#): Plan IGN map backgrounds.

[IGN RGE ALTI](#): local terrain windows and derived crests.

Licence Ouverte 2.0; only the selected cells downloaded.

Locator and analysis

[Natural Earth](#): France outline, public domain.

MSD, intervals and crossings: this thesis's prepared pipeline.

[Code, figure scripts and recorded source provenance](#)

Vilpellet, Darmon & Benzaquen (2026). *From Random Walks to Thermal Rides: Universal Anomalous Transport in Soaring Flights*. [arXiv:2601.01293](https://arxiv.org/abs/2601.01293).

Transport, HMM phase segmentation and phase-duration benchmarks.

NIST/SEMATECH. *e-Handbook of Statistical Methods*,
[§5.2.1.2: One Variable at a Time](#); [§5.3.3.3: Full Factorial Designs](#).

Mohr, Wilson & Freund (2022). *Statistical Methods*, 4th ed., §9.3, §9.7.

FAA (2024). *Glider Flying Handbook*, Ch. 9–10.

Soaring Society of America. *Lift Sources*.

Schroeder & Buck (1970). *Fire Weather*, Ch. 6: General Winds.

Savitzky & Golay (1964). Smoothing and Differentiation of Data by Simplified Least Squares Procedures. *Analytical Chemistry* 36, 1627–1639. [doi:10.1021/ac60214a047](https://doi.org/10.1021/ac60214a047).

What the data show

Persistent displacement growth.

A fixed population makes lag comparisons interpretable; H summarises one fit range.

Circuit and terrain interact.

Open > closed within every altitude class.
The altitude ordering reverses by circuit.

Composition matters.

Group MSD shares change with lag;
flight counts alone cannot explain pooled H .

What remains open

Mechanisms and uncertainty.

Terrain-linked lift is observed; wind and route mechanisms need matched tests.

Cadence and scaling.

Validate segmentation, anisotropy, distribution scaling and bootstrap coverage.

Joint models and social flight.

Quantify conditional group / solo contrasts;
test a phase-based stochastic model.

Next: test the mechanisms with joint models and validated flight phases

Thank you

Questions and discussion

Thanks to the pilots sharing their flights,
the FFVL archive and the research community.

A: 4 altitudes C: 2 circuits E: beginners / experts $4 \times 2 \times 2 = 16$ cells

$$H_{ace} = \underbrace{\mu + A_a + C_c + E_e}_{\text{additive part}}$$

$$+ (AC)_{ac} + (AE)_{ae} + (CE)_{ce} + (ACE)_{ace}$$

Example: two altitude levels and two circuits; beginners / experts group held fixed.

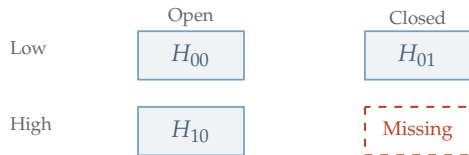
Free coefficients:

6 additive

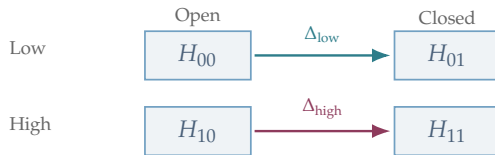
10 interactions

Sum-to-zero constraints

OFAT: one combination missing



Full factorial: all four combinations

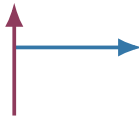


Interaction: $D_{AC} = \Delta_{\text{high}} - \Delta_{\text{low}} \neq 0$ means the circuit effect depends on altitude

$$\mathbf{v}(t) = s(t) \mathbf{e}(t), \quad \mathbf{e} = (\cos \theta, \sin \theta) \quad \Delta\theta = \theta(t + \tau) - \theta(t)$$



Aligned: $\cos \Delta\theta = 1$



Perpendicular: 0



Opposite: -1

$$C_v(\tau) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle}{\langle s^2 \rangle} = \frac{\langle s(t)s(t + \tau) \cos \Delta\theta \rangle}{\langle s^2 \rangle}$$

$$C_\theta(\tau) = \langle \mathbf{e}(t) \cdot \mathbf{e}(t + \tau) \rangle = \langle \cos \Delta\theta \rangle$$

Constant speed: $C_v = C_\theta$.

Persistent mean drift can leave a nonzero baseline; also inspect centred velocity covariance.

C_v combines speed and direction; C_θ isolates directional agreement

1 Group flights

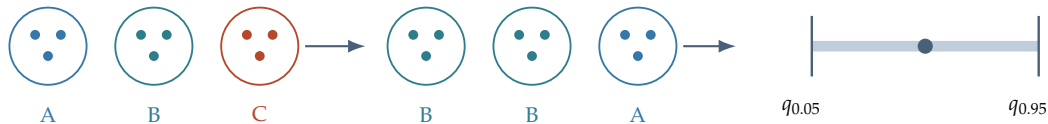
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined